

Improving the Response Choices on the Veterans SF-36 Health Survey Role Functioning Scales

Results From the Veterans Health Study

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Abstract: Role functioning and its limitations due to one's health is an important aspect of health-related quality of life (HRQoL). The Medical Outcomes Study (MOS) SF-36 includes 2 role functioning scales: role limitations due to physical health problems (RP) or emotional problems (RE). Although they capture important concepts of HRQoL, these 2 scales have some limitations in their measurement properties. Using dichotomized sets of response choices, the scales are limited in their distributional properties (eg, higher standard deviation than other SF-36 scales) and ability to discriminate between clinically relevant groups. In this study, we ascertain the improvements to these 2 scales using 5-point ordinal response choices for each of the scale items. Two thousand one hundred sixty-two patients from the Veterans Health Study (VHS), an observational study of health outcomes in patients receiving ambulatory care, completed a health status questionnaire and a medical history. The health questionnaire included (1) the MOS SF-36, in which the RP and RE items used dichotomized yes/no responses; and (2) a set of modified RP and RE items that used 5-response choices for each of the items, ranging from "no, none of the time" to "yes, all of the time." We compared the original and modified RP and RE scales using internal consistency reliability and factor analysis. We tested item convergent and discriminant validity using multitrait scaling, and scale discriminant validity using ordinary least squares regression. Results indicate that the modifications to the original RP and RE scales accomplish important gains in the distributional properties of the scales. The floor and ceiling effects of the 2 scales have been reduced and the reliability of the RP scale has increased (0.87-0.95). Factor analysis and multitrait scaling tests indicate

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This study was supported by grant SDR 91006.S from the Health Services Research and Development Service Department of Veterans Affairs, Washington, DC.

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that the modified items have the same interpretation as the original items. Tests of discriminant validity indicate that the modified RP and RE scales have greater explanatory power for measures of disease burden, depression, and disease severity. The modified SF-36 role scales are clearly superior to the original versions. The modifications have increased the explained variability, suggesting greater explanatory power and more information obtained by the role functioning measures. The modified RP and RE are capturing a wider spectrum of disease severity, in part due to the lowering of the floor and raising of the ceiling of the scales. Additional work needs to test these improvements in other populations and to expand the analysis to track the responsiveness of the modified scales to clinically and socially important changes over time. **Key words:** *health outcomes, health status, Veterans SF-36*

ROLE FUNCTIONING is described as the ability or capacity of an individual to perform certain activities at home, work, or at school. Role functioning is often related not only to one's own expectations and satisfaction that one might derive in the fulfillment of one's roles in society, but also to the individual's social and cultural environment and prescribed societal behavioral patterns that are often circumscribed by existing institutions, be they in the workplace, community, or at home (Derogatis et al., 1974). Because individuals in our society often function in different roles, the ability or capacity of the individual to perform these roles thus consists of an important component of health-related quality of life (HRQoL). The roles more commonly described include structured work (that is, work for salary), work at home, performance of household chores, as well as child care and one's role as a student (Nagi, 1969).

More recently, patient-centered measures of outcome have been developed. These measures have been applied to the assessment of the functional status (physical, psychological, social, and role) or disease burden of patient populations and as measures of the outcomes of healthcare. The Medical Outcomes Study (MOS) building on this paradigm sought to integrate the limitations in one's health with the concept of role functioning, as part of a conceptual approach to defining role within an overall model of HRQoL (Tarlov et al., 1989). Defining role functioning in terms of health limitations helps to focus this concept within the domain of health outcomes (Levine & Croog, 1985; Stewart et al., 1978). This model distinguishes between physical and mental

health and is a 2-dimensional model, where other aspects of health fit within or between these 2 dimensions. This dichotomy has been well documented and provides a rubric for assessing role functioning by using limitations due to physical and mental health problems. The MOS expanded the definition of the role concept through the inclusion of a broad range of role functioning due to health limitations and a summated scale based upon a set of dichotomized items. This set of items was used as a continuous scale. Originally, a longer single set of items for role functioning was included. This was split out into a final set of 2 concepts, 4 role items related to physical problems (role-physical [RP]) and 3 role items related to emotional problems (role-emotional [RE]). This was included as 2 separate concepts or scales in the MOS SF-36 (short form health survey), a general measure of HRQoL.

However, role functioning in health outcome measures has often been conceptualized as an all-or-none phenomenon, in terms of performance of one's work or other activities, quality of work, or time spent on work. In the MOS SF-36, the 2 role limitations scales (due to physical or emotional problems) were constructed to provide dichotomized response choices at the item level (eg, yes versus no) (Ware & Sherbourne, 1992). Summation across items gives a continuous scale with some major limitations. First, the scale is coarse since it provides a more limited set of possible scores from 0 to 100 than do other scales in the SF-36. There are 5 possible scores from 0 to 100 on the 4-item RP scale and 4 possible scores on the 3-item RE scale. The small number of possible scale scores result in

distributional problems related to substantial proportions of patients who may rest at 0 or 100 on the 2 scales, also termed *ceiling* and *floor effects*. Second, there are some limitations in precision or reliability that might otherwise be gained with additional response choices added for each of the items. These gains in precision may have implications for the ability to discriminate between clinically relevant patient groups. Third, there may be loss of information if a dichotomized set of items is used instead of a set of ordinal choices for each item, and therefore a lack of explanatory power in the relationship between disease status and role functioning. Fourth, the coarseness of the scale may miss clinically relevant associations because they are not statistically significant.

The purpose of this article is to test the reliability and validity of a modified set of items by using multiple response choices for the RP and RE scales contained in the SF-36. The comparisons use both sets of scales, that is, the original dichotomized item set and a revised 5-point set of response choices for each of the items. We compared the 2 sets of scales in a veteran ambulatory population of Veterans Administration (VA) users of care in the Veterans Health Study (VHS) (Kazis et al., 1996, 1998). Comparisons involved assessing changes in the distributional properties of the scales. We also compared gains in precision or reliability. We examined the strength of associations with clinically based measures of symptom severity in specific subgroups of patients with established chronic disease as a measure of the concurrent validity of the measures. Finally, we discuss conversion from the modified role scales to the dichotomized version for comparison with other studies.

METHODS

Data for this study are from the VHS, an observational study of health outcomes in patients receiving VA ambulatory care (Kazis et al., 1999). The VHS recruited patients who used ambulatory care services at 1 of 4 VA

outpatient clinics in the greater Boston area between June 1993 and March 1996. Patients were approached in clinic by research staff during specified periods of time, and a total of 6829 patients who had at least 1 VA visit in the past year completed a brief screening questionnaire and provided contact information. From this pool of potential study participants, 5555 were selected randomly, and 4236 were considered eligible and were invited into the study. Of these eligible patients, 2425 (57%) agreed to participate and were mailed a health status questionnaire (including the SF-36) and were scheduled for an in-person visit within a month. Two thousand one hundred sixty-two patients (89%) completed the health status questionnaire and visit, and had complete data. Participating patients were similar to other clinic patients in terms of general health status, disease profiles, and sociodemographic characteristics, except that they had slightly more education (12.2 years for participants versus 11.9 for others). The VHS identified a number of study conditions based on a patient's report of having a physician's diagnosis, treatment, symptoms, or all of these. Patients identified with each of these conditions were asked a separate set of questions for a more in-depth review of their current symptoms, treatment, or both. This study focuses on some of those conditions, which include diabetes (DIAB), chronic lung disease (CLD), low back pain (LBP), and osteoarthritis of the knee (OA). At the time of the in-person interview, a medical history questionnaire was also administered.

STUDY MEASURES

SF-36 (Short Form Health Survey)

The SF-36 is one of the measures of HRQoL in the VHS study (Ware & Sherbourne, 1992). The MOS SF-36 is a self-reported health study measuring 8 different concepts of HRQoL. These include physical functioning (PF), role limitations due to physical problems (RP), bodily pain (BP), general health perceptions (GH), vitality (VT), social functioning (SF), role limitations due to

emotional problems (RE), and mental health (MH). This assessment has been extensively documented as reliable and valid in ambulatory patient populations. Items from each scale are summed and rescaled with a standard range from 0 to 100, where 100 denotes the best health. In the present study, we examined role limitations due to physical problems (RP) and role limitations due to emotional problems (RE). These scales were the focus of this article because of limitations in their distributional properties, as well as some limitations in their precision or reliability.

The MOS SF-36 was collected as part of the initial mailed survey. In addition, we included a separate set of modified questions for role limitations due to physical and emotional problems (RP and RE) separated by another set of questions. The modified RP scale is measured by asking "During the past 4 weeks, have you had any of the following problems with your work or other regular daily activities as a result of your physical health?" The questionnaire provides 4 items, each with the same set of 5 response choices. The items include the following: (1) cut down the amount of time you spent on work or other activities, (2) accomplished less than you would like, (3) were limited in the kind of work or other activities, or (4) had difficulty performing the work or other activities (eg, it took extra effort). The questionnaire provides 5 response categories for each item (*No, none of the time; Yes, a little of the time; Yes, some of the time; Yes, most of the time; and Yes, all of the time*). Other than the change in response choices, there are no modifications to the RP scale when compared to the original version from the MOS SF-36, which gives a dichotomized set of yes-or-no choices for each item. The RE scale asks "During the past 4 weeks, have you had any of the following problems with your work or other regular daily activities as a result of any emotional problems (such as feeling depressed or anxious)?" There are 3 items that ask, (1) "Cut down the amount of time you spent on work or other activities," (2) "Accomplished less than you would like," and (3) "Didn't do work or other activities as carefully as usual." The

same 5 response categories are given as in RP. We use RP2 and RE2 to indicate the original scales with dichotomized choices, and RP5 and RE5 for the scales using 5-point ordinal choices. Both versions of the role scales were fielded in the same questionnaire.

Disease severity

Disease severity is measured for each of 4 conditions under study using symptoms or treatment pertaining to each condition. Detailed descriptions of these disease severity measures have been described elsewhere and are summarized briefly (Clark et al., 1998; Fincke et al., in press; Selim et al., 1997, 1998). The severity measure of DIAB includes a variable list that corresponds to the well-known chronic complications of DIAB. These include eye disease, foot disease, autonomic neuropathy, peripheral neuropathy, atherosclerosis (including any history of stroke, TIA or MI; chest pain frequency; and claudication). The severity measure of CLD includes dyspnea and intensity, wheezing frequency and intensity, frequency of coughing, and amount of speed and production. The severity of LBP measures pain intensity and frequency. The severity measure of OA includes pain with activity, pain at rest, impaired mobility, and sensations of crepitus. Each of the symptom-based severity measures range from 0 (least severe) to 100 (most severe), and has been shown to be a reliable and valid instrument. The study subjects had a mean severity score of 30.3 (SD = 22.8) for DIAB, 53.3 (SD = 18.3) for CLD, 51.1 (SD = 21.1) for LBP, and 46.5 (SD = 21.1) for OA. All severity scores have been transformed to a normal standard from 0 to 100, where 100 denotes severe disease.

Medical history

Medical history was assessed using an interview or administered questionnaire previously adapted and modified from the MOS. This questionnaire has been clinically validated and tested in prior work (Selim et al., 2002). Using information obtained from the medical history, we developed an index of the number of health conditions or morbidities

(also called a Disease Burden Index [DBI]) that range from 1 to an observed maximum of 22. These health conditions include symptoms and reported morbidities in addition to those identified and targeted in the VHS (eg, anemia, angina pectoris, congestive heart failure, dermatitis, diverticulitis, enlarged prostate, dizzy spells, fainting spells, gout, inflammatory bowel disease, hypertension, skin cancer, and thyroid disease).

A separate index was constructed using the 20-item index of the CESD (Center for Epidemiologic Studies in Depression) (Radloff, 1977). The CESD index is a self-reported measure of depressive symptoms. The medical and CESD index have been validated against SF-36 and measures of utilization in prior work (Kazis et al., 1998).

Sociodemographics

Sociodemographic information reported includes age (calculated using the date of birth and the date of initial contact in the study), years of education (range of 1 to more than 16 years), race, marital status, and income (less than \$20,000).

ANALYSES

We compared internal consistency reliability of the original and modified RP and RE scales using the Cronbach alpha statistic. We conducted factor analyses to determine if the relationship between the RP and RE concepts at the item and scale levels with other SF-36 concepts was comparable using the original and modified versions. Separate factor analyses were performed on the 35 items (omitting the change item) using the original and modified RP and RE items. Factor analyses were then conducted on the 8 SF-36 scales, also using the original and modified scales. All factor analyses used principal components with varimax rotation.

To test the performance of items in their hypothesized scales, multitrait scaling was conducted to ascertain item-scaling correlations, or convergent and discriminant validity for both the original and modified RP and RE (Hays & Hayashi, 1990). Item internal con-

sistency is assessed by determining whether each item in a scale is substantially linearly related to the total score computed from other items in that same scale. The item discriminant validity criterion is assessed by determining whether each item has high correlations with the scale it is hypothesized to belong to than with all other scales. Item internal consistency was supported if an item correlated substantially ($r \geq 0.40$) with the scale it was hypothesized to represent. To correct for overlap, the hypothesized item is deleted from the scale with which it is correlated. Item discriminant validity is dependent upon the magnitude of the correlation between an item and its scale relative to the correlation of that item with other scales. Hypothesized correlations more than 2 standard errors higher than the other correlations were considered a scaling success; those within or less than 2 standard errors lower were considered a scaling error.

We compared the original and modified RP and RE scales using descriptive statistics. These included the mean and standard deviation, the coefficient of variation, and the percentage of patients at the floor of the scale (0) and at the ceiling (100). We were interested in determining whether there was a reduction in ceiling and floor effects in the revised scales. We also examined those at the 25th, 50th, and 75th percentiles of the distribution for the original and modified scales.

To examine discriminant validity, we used ordinary least squares (OLS) regression to ascertain the relationship or association between symptom-based disease severity and the original and modified RP and RE scales. Because of the hypothesized increase in precision for the modified scales, we would expect that the regression models would have higher R^2 values than would the original scales. This is because greater precision should reduce the variance of the scores and provide tighter confidence intervals, and therefore increase the statistical power to detect significant differences.

For most analyses, the modified RP and RE measures were scored in the traditional way. That is, the average was formed for all

items using a 100, 75, 50, 25, and 0 rescaling of the item responses, provided that more than half of the responses were present. We also calculated a method for conversion of the 5-response format to the dichotomized 2-response format for each of the scales. Using cases in which both 2-response and 5-response versions were available in the same questionnaire, we calculated the probability of a yes response in the yes/no original format for each response in the 5-response modified format and multiplied this by 100. This resulted in a recoded score for each response. We averaged the scores for each item. To account for regression-to-the-mean bias, we matched the 5-response scales to the corresponding 2-response scales in terms of mean and standard deviation.

Last, we computed a measure of the true difference between the original and modified RP and RE scales. This involved computing the variance of the difference of 2 and then subtracting from it the variance attributable to the unreliability of the measurement.

$$\sigma_{\Delta}^2 - [(1 - \alpha_{R2})\sigma_{R2}^2 + (1 - \alpha_{R5})\sigma_{R5}^2]$$

where σ_{Δ}^2 is the variance of the difference, σ_{R2}^2 and σ_{R5}^2 are the variances of the 2-response and 5-response variances, and α_{R2} and α_{R5} are the respective reliabilities.

We hypothesized that we would see little in the way of real differences between the original and modified scales; that is, the standard deviation of the true differences should be close to 0.

RESULTS

In the sample of 2424 VA ambulatory patients, ages ranged from 22 to 90 (mean = 62, median = 65). Fifty-eight percent of the patients were married, 92% patients were almost exclusively white, and 52% had an income less than \$20,000/year.

Patients had an average of 6.0 diagnoses/symptoms, indicating substantial comorbidity in this population. The most prevalent conditions were angina pectoris (27%), LBP (24%), OA (21%), DIAB (17%), and CLD (14%). Twenty-eight percent were screened positive for depression, and 12% with alcohol-related problems (Table 1).

Table 2 presents descriptive statistics of the SF-36 using the 8 scales without modification to the RP and RE concepts. The RP and RE scales describe the greatest variation as indicated by the standard deviation and coefficient of variation. Standard deviations for the 2 scales were above 41 and 43 compared to the other scales for which these ranged from

Table 1. Description of Veterans Health Study (VHS) population

Variable	N (%)	Mean	SD	Min	Max
Age, y	2156	61.8	12.6	22	90
Education, y	2158	12.6	2.5	3	17
Disease Burden Index (40 items)	2162	6.0	3.8	0	25
Married	1253 (58)				
White	1986 (92)				
Income <\$20,000	998 (52)				
Angina	582 (27)				
Diabetes	374 (17)				
Chronic lung disease	312 (14)				
Osteoarthritis	449 (21)				
Low back pain	513 (24)				
Cancer	292 (14)				
Alcohol-related problems	260 (12)				
Depression	631 (28)				

Table 2. SF-36 self-reported health status ($N = 2162$)

SF-36 scale	N	Mean	SD	CV*	% Floor (0)	% Ceiling (100)
Physical functioning (PF)	2153	57.2	28.7	50.2	1.5	6.2
Role physical (RP)	2162	38.2	41.0	107.3	44.7	22.7
Bodily pain (BP)	2158	55.3	26.9	48.7	2.3	11.8
General health perceptions (GH)	2151	51.2	23.2	45.3	0.9	0.9
Vitality (VT)	2160	52.5	23.3	44.3	2.1	0.7
Social functioning (SF)	2162	67.4	28.1	41.7	2.4	26.3
Role emotional (RE)	2162	61.1	43.1	70.5	26.9	49.7
Mental health (MH)	2157	69.0	22.7	32.9	0.6	4.1

*CV indicates coefficient of variation.

23 to about 29. We also found that the percentage of patients at the floor of the scale (a score of 0, worst health) was highest for these 2 scales. The percentage of patients that was at the ceiling of the distribution (100 that denotes best health) was also among the highest for these 2 scales with the exception of social functioning.

Table 3 provides the descriptive statistics for RP and RE for the original and modified versions. The Cronbach alpha statistics, an indication of the internal consistency and reliability, has improved for the RP in the modified version, while it is comparable for the RE scale. Direct comparisons of the descrip-

tive statistics indicate some reduction in the standard deviation and coefficient of variation. More interestingly, the percentage of patients at the ceiling of the distribution have been reduced by about 9.5% for the modified RP and 23% for the modified RE. The distribution of those at the floor has also been markedly reduced by about 38% for RP and 25% for RE. The percentile distribution is also shifted in the modified RP and RE scales. Dramatic shifts for RP are noted at the 25th and 50th percentiles, with a difference of 25% for each, while for RE, differences of 50% and 8% are also noted at the same percentiles. For RP, the skewness and kurtosis have been

Table 3. Descriptive statistics for role-physical and role-emotional scales

	Role-physical		Role-emotional	
	Original	Modified	Original	Modified
Cronbach α	.87	.95	.87	.88
Mean	38.24	53.07	61.11	70.98
Standard deviation	41.04	31.26	43.10	26.79
Coefficient of variation	107.33	58.89	70.52	37.75
Range	100.00	100.00	100.00	100.00
% Ceiling	22.71	13.18	49.72	26.36
% Floor	44.73	7.12	26.87	2.22
Skewness	0.48	-0.01	-0.43	-0.71
Kurtosis	-1.43	-1.13	-1.56	-0.33
25th Percentile	0	25	0	50
Median	25	50	67	75
75th Percentile	75	81	100	100

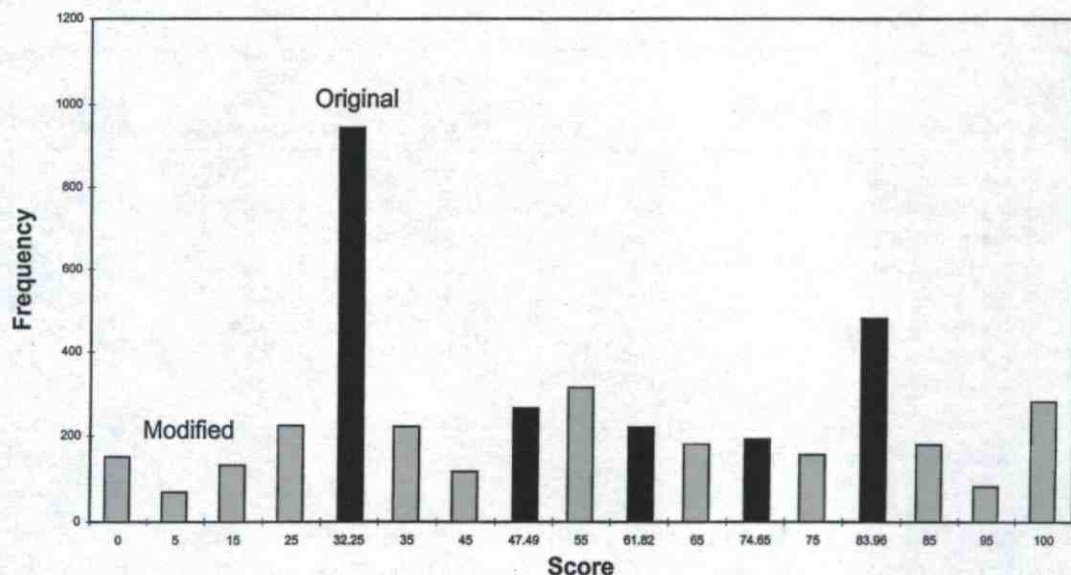


Figure 1. Frequency distributions of the role-physical scale: original and modified.

reduced, while for RE the skewness continues to be somewhat marked and kurtosis is reduced.

Figures 1 and 2 show the frequency distributions of the 2 scales comparing the original and the modified versions of RP and RE respectively. In these diagrams, the original

values are plotted at the mean value of the revised scale. The distributions indicate a metric of 17 units for the modified version, and only 5 for the original RP scale (Fig 1); 15 units for the modified RE scale, and only 4 for the original scale (Fig 2). Both figures indicate that there has been a lowering of the

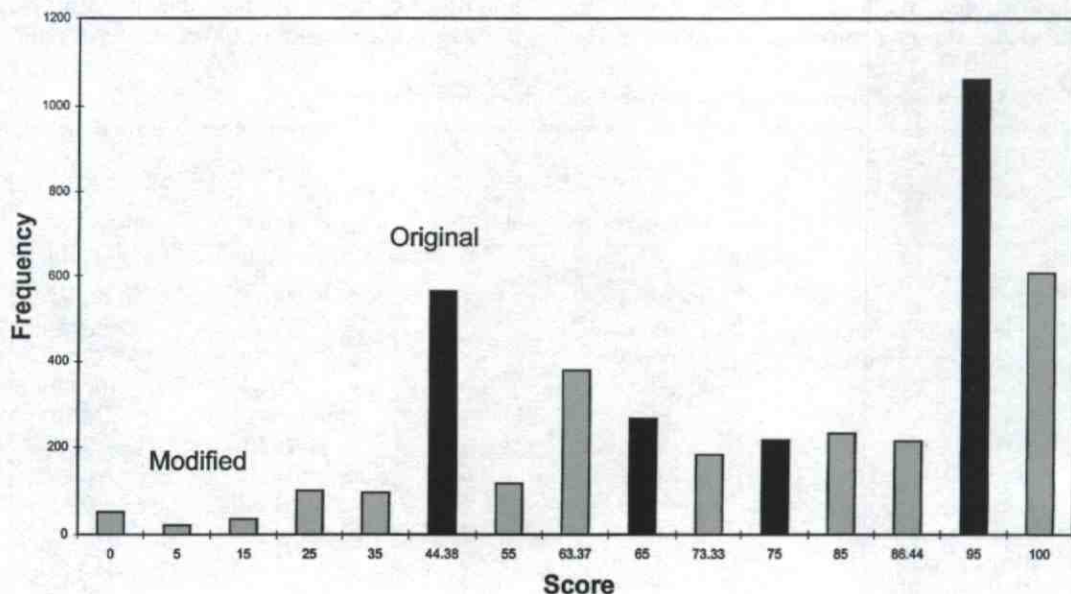


Figure 2. Frequency distributions of the role-emotional scale: original and modified.

Table 4. Results of item scaling tests for SF-36 scales

Scale	No. of items	Item internal consistency*	Item discriminant validity†	No. of successes/total‡	Scaling success rate, %
Original role items					
Physical functioning	10	0.54-0.81	0.20-0.60	70/70	100.0
Role physical	4	0.71-0.74	0.26-0.56	28/28	100.0
Bodily pain	2	0.76	0.28-0.57	14/14	100.0
General health	5	0.53-0.74	0.34-0.53	35/35	100.0
Vitality	4	0.71-0.74	0.32-0.80	28/28	100.0
Social functioning	2	0.66	0.35-0.59	14/14	100.0
Role emotional	3	0.71-0.78	0.18-0.58	21/21	100.0
Mental health	5	0.64-0.79	0.18-0.58	35/35	100.0
Total				245/245	100.00
Modified role items					
Physical functioning	10	0.54-0.81	0.17-0.68	69/70	98.6
Role physical	4	0.86-0.89	0.28-0.68	28/28	100.0
Bodily pain	2	0.76	0.27-0.62	14/14	100.0
General health	5	0.52-0.73	0.34-0.58	35/35	100.0
Vitality	4	0.71-0.75	0.32-0.58	28/28	100.0
Social functioning	2	0.66	0.32-0.62	13/14	92.9
Role emotional	3	0.70-0.80	0.24-0.58	21/21	100.0
Mental health	5	0.63-0.79	0.18-0.60	35/35	100.0
Total				243/245	99.2

*Correlations between items and hypothesized scales corrected for overlap.

†Correlations between items and other scales.

‡Number of hypothesized items significantly higher (2 standard errors)/total number of correlations.

floor and a raising of the ceiling when the modified version is compared to the original RP and RE scales. In Figure 1, those who scored at 0 in the modified RP scale scored at 32.25 in the original, while those who scored 100 in the modified scale scored at 83.96 in the original. In Figure 2, those who scored at 0 and 100 for the modified RE, scored at 44.38 and 95, respectively, for the original version.

Table 4 shows results of multitrait item scaling tests. Item internal consistency is supported for the original and modified role items ($r \geq 0.40$). Item discriminant validity was strongly supported for the original items, with 245 scaling successes out of 245. For the multitrait scaling tests that included the modified role items, there were 243 scaling successes out of 245 tests (99.2%). With the exception of the 2 correlations that are within

1 standard error higher of the RP scale, with one physical function item (vigorous activities) and one social functioning item (interference with normal social activities), all other tests were more than 2 standard errors higher than the correlations with other items.

Tables 5 and 6 present the factor loadings for the 35 items of the SF-36 using the original role scale items and the modified items. Results indicate that the modified role scale items are distinct from the other factors or scale concepts and, for the most part, relate in a similar fashion to the other factors when compared with the factor pattern that used the original role scale items. The communality estimates for the 4 RP items are higher for the modified role-scale items. The RE items are also slightly higher for the modified version. The cumulative percentage

Table 5. Factor loadings obtained from factor analysis of the 35 items using original role scales (Varimax method)

Subscales and items	Factors								Communality
	I	II	III	IV	V	VI	VII	VIII	
Physical functioning									
Vigorous activities	52								0.64
Moderate activities	69								0.68
Lifting or carrying groceries	71								0.65
Climbing several flights of stairs	74								0.69
Climbing one flight of stairs	80								0.71
Bending, kneeling, or stooping	64								0.59
Walking more than a mile	82								0.76
Walking several blocks	86								0.80
Walking one block	81								0.73
Bathing or dressing yourself	52								0.59
Role physical									
Cut down amount of time at work			70						0.69
Accomplished less than would like			76						0.73
Limited in kind of work			73						0.73
Difficulty performing work			73						0.72
Bodily pain									
Bodily Pain							72		0.77
Pain interfering with normal work							72		0.79
General health									
General health	40			58					0.67
Get sick a little easier				62					0.61
As healthy as anybody				69					0.68
Expect health to get worse				68					0.57
Excellent health				68					0.75
Vitality									
Full of pep					70				0.75
Lots of energy					70				0.79
Feel worn out					65				0.78
Feel tired					71				0.78
Social functioning									
How much interfered with social							48		0.66
How often interfered with social							44		0.60
Role emotional									
Cut down amount of time at work						77			0.78
Accomplished less than would like						80			0.80
Didn't do work as carefully						75			0.71
Mental health									
Very nervous		81							0.72
Down in the dumps		79							0.77
Calm and peaceful		61							0.68
Downhearted and blue		76							0.77
Happy		58							0.69
Variance	6.42	3.66	3.18	2.98	2.77	2.63	2.26	0.92	
% Proportion	18.34	10.45	9.09	8.52	7.92	7.52	6.45	2.64	
% Cumulative	18.34	28.79	37.88	46.40	54.32	61.84	68.29	70.93	

Table 6. Factor loadings obtained from factor analysis of the 35 items using modified role scales (Varimax method)

Subscales and items	Factors								Communality
	I	II	III	IV	V	VI	VII	VIII	
Physical functioning									
Vigorous activities	51							-40	0.67
Moderate activities	65								0.69
Lifting or carrying groceries	68								0.64
Climbing several flights of stairs	74								0.70
Climbing one flight of stairs	79								0.72
Bending, kneeling, or stooping	64								0.64
Walking more than a mile	81								0.76
Walking several blocks	85								0.80
Walking one block	79								0.76
Bathing or dressing yourself	48							59	0.65
Role physical									
Cut down amount of time at work			73						0.84
Accomplished less than would like			76						0.86
Limited in kind of work			76						0.85
Difficulty performing work	41		71						0.85
Bodily pain									
Bodily Pain							77		0.84
Pain interfering with normal work							72		0.83
General health									
General Health				59					0.66
Get sick a little easier				62					0.57
As healthy as anybody				69					0.68
Expect health to get worse				69					0.55
Excellent health				69					0.73
Vitality									
Full of pep					67				0.73
Lots of energy					68				0.78
Feel worn out					66				0.75
Feel tired					72				0.76
Social functioning									
How much interfered with social		40	43						0.68
How often interfered with social		44							0.62
Role emotional									
Cut down amount of time at work						76			0.82
Accomplished less than would like						77			0.83
Didn't do work as carefully						75			0.74
Mental health									
Very nervous		74							0.62
Down in the dumps		77							0.74
Calm and peaceful		74							0.67
Downhearted and blue		78							0.76
Happy		74							0.68
Variance	6.40	4.27	3.77	3.15	2.63	2.34	1.85	1.04	
% Proportion	18.29	12.20	10.76	9.01	7.53	6.68	5.27	2.97	
% Cumulative	18.29	30.49	41.25	50.26	57.79	64.47	69.74	72.71	

Table 7. Factor loadings obtained from the factor analysis of the SF-36 scales

Scale	Factor		Communality
	I	II	
Using original role scales			
Physical functioning	86		0.76
Role physical	74		0.64
Bodily pain	79		0.67
General health	72		0.65
Vitality	51	65	0.67
Social functioning	61	57	0.69
Role emotional		81	0.70
Mental health		89	0.83
Variance explained	3.12	2.50	
% Proportion	39.01	31.23	
% Cumulative	39.01	70.24	
Using modified role scales			
Physical functioning	88		0.79
Role physical	83		0.79
Bodily pain	78		0.67
General health	67	43	0.64
Vitality	45	70	0.70
Social functioning	59	59	0.70
Role emotional		80	0.71
Mental health		91	0.84
Variance explained	3.16	2.67	
% Proportion	39.56	33.39	
% Cumulative	39.56	72.94	

explained by the factor structure is slightly higher when using the modified role scale items.

Table 7 presents results of the factor analyses using the 8 SF-36 scales. Results indicate comparable factor structures with factor 1 being a physical factor and factor 2 being a psychologic factor. Communality estimates for RP are higher for the factor structure using the modified role scale. RE is comparable for the 2 factor structures. The cumulative proportion explained by the factor structure is slightly higher for the one using the modified role scales.

Results of the regression models for the overall DBI and the symptom-based measure for depression (CESD) are shown in Table 8. Results indicate an increased R^2 for the mod-

ified RP and RE when compared with the original version for both the DBI and the CESD.

Table 9 presents the results of the regression of disease severity on the RP and RE scales using the original and modified versions. Results indicate increases in explanatory power as reflected by the R^2 for CLD, OA, LBP, and, to a lesser extent, DIAB. For role emotional, the modified scale shows increased explanatory power for CLD and LBP.

Using the method for computing the true difference between the modified and original RP and RE scales, we found that the standard deviation of the true difference was 27.9 for RP and 30.1 for RE. The algorithm for converting the modified RP and RE back to the original is given in the Appendix.

Table 8. Regression of overall Disease Burden Index and CESD on RP and RE SF-36 scales: original and modified versions*

Dependent variable	Independent variable	N	Coefficient	SE	Standard coefficient	R ² , %	P [†]
Role physical							
RP	DBI	2178	-4.55	0.21	-21.44	17.4	.0011
RP5	DBI	2178	-3.68	0.16	-23.13	19.7	
RP	CESD	2165	-1.35	0.07	-18.30	13.4	.0193
RP5	CESD	2165	-1.13	0.06	-20.56	16.3	
Role emotional							
RE	DBI	2178	-4.20	0.23	-18.45	13.5	.00001
RE5	DBI	2178	-2.90	0.14	-20.88	16.7	
RE	CESD	2165	-2.31	0.07	-34.73	35.8	.00001
RE5	CESD	2165	-1.52	0.04	-38.17	40.3	

*DBI indicates Disease Burden Index; CESD, Center for Epidemiologic Studies in Depression; RP, role-physical (original); RP5, role-physical (modified); RE, role-emotional (original); and RE5, role-emotional (modified).

[†]For testing the difference between 2 regression coefficients.

Table 9. Regression of disease severity on RP and RE SF-36 scales: original and modified versions*

Dependent variable	Independent variable	N	Coefficient	SE	Standard coefficient	R ² , %	P [†]
Role physical							
RP	CLD severity	312	-0.52	0.10	-5.15	7.9	.9513
RP5	CLD severity	312	-0.53	0.08	-6.45	11.8	
RP	OA severity	373	-1.51	0.18	-8.51	16.3	.2991
RP5	OA severity	373	-1.28	0.14	-9.36	19.1	
RP	LBP severity	477	-0.48	0.07	-7.37	10.3	.3288
RP5	LBP severity	477	-0.56	0.05	-10.79	19.7	
RP	DIAB severity	341	-0.66	0.11	-5.81	9.1	.3751
RP5	DIAB severity	341	-0.54	0.09	-6.17	10.1	
Role emotional							
RE	CLD severity	312	-0.48	0.13	-3.71	4.3	.8199
RE5	CLD severity	312	-0.45	0.08	-5.42	8.7	
RE	OA severity	373	-1.52	0.22	-7.02	11.7	.027
RE5	OA severity	373	-0.95	0.14	-6.94	11.5	
RE	LBP severity	477	-0.62	0.09	-6.95	9.2	.1331
RE5	LBP severity	477	-0.46	0.06	-8.27	12.6	
RE	DIAB severity	341	-0.54	0.12	-4.31	5.2	.0517
RE5	DIAB severity	341	-0.26	0.07	-3.49	3.5	

*RP indicates role-physical (original); RP5, role-physical (modified); RE, role-emotional (original); RE5, role-emotional (modified); CLD, chronic lung disease; OA, osteoarthritis of the knee; LBP, low back pain; and DIAB, diabetes.

[†]For testing the difference between 2 regression coefficients.

DISCUSSION

Results of this study indicate that the modifications to the RP and RE scales provided important gains in almost all tests. The floor and ceiling effects of the 2 scales have been reduced, and the reliability of the role physical scale has been increased. The addition of more increments in the modified scales (from 5 to 17 for RP, and from 4 to 15 for RE) has the effect of increasing the precision of the scale, with positive impacts on the distributional properties of the scales, including a raising of the ceiling and a lowering of the floor. The skewness of the modified RP scale is also much less pronounced.

The results of the factor analysis indicate a factor structure that is relatively unchanged; thus, the modified role items and scales have the same interpretation as the original ones. This suggests that the relationship between the role concept and the other SF-36 concepts is relatively unchanged or consistent. The modified version provides more information. The modified scales had higher communality estimates for the RP and RE items, and this explained more of the variability for this factor structure. Multitrait testing indicated that the discriminant validity of the modified scales was also consistent with the original version.

Tests of concurrent validity by the modified version, using a DBI and a measure of depressive symptoms, were quite strong. Results indicated that the proportion of explained variability increased for both the DBI and CESD for the RP and RE scales. Similarly, for the modified RP scale, the symptoms of disease severity showed greater explanatory power for CLD, OA, LBP, and, to a lesser extent, DIAB. For RE, not surprisingly, increases in explained variability were found for CLD and LBP symptom severity measures. Both lung disease and LBP have well-documented emotional impacts on health.

The increases in explained variability suggest greater explanatory power and more information explained by the role functioning measures in relationship to disease severity indicators. Role functioning is capturing a wider

spectrum of disease severity, perhaps because of the lowering of the floor and raising of the ceiling of the scale. In addition, there are increases in precision that are likely to give more information because of greater discriminatory power through more units in the metric defining the modified RP and RE scales. The standard error has been reduced, given no change in sample size, from 0.88 to 0.67 for RP, original and modified, respectively; and from 0.92 to 0.58 for RE. The modified scales have narrower confidence intervals and consequently smaller sample size requirements to detect significant differences between clinical groups under study (a savings in efficiency).

Interestingly, the measure of the true difference between the original and modified scales for the RP and RE scales were not 0. There are several possible explanations for this. First, there is the strong possibility that because of intervening questions located between the original and modified role items, patients may have been influenced in their responses. A contextual effect may have been present. Tests conducted on other response sets indicate that differences of this magnitude have been found where interim questions impact on scores (Miller et al., 2004). Second, because of the formatting of the role items where all are scored by the respondent in the same way and direction, patients may not have read each item in a careful fashion, and the internal consistency reliability coefficient may have been somewhat inflated by the list-format responses, adding another source of error. Third, there is the possibility that the changes to the items result in a change in the concept or domain. This is unlikely to be a major effect, given that the relationship between the 2 modified role concepts and the other SF-36 concepts is consistent with the associations found in the original role concepts using results of the factor analysis and discriminant validity.

This study had several limitations. First, this study was cross-sectional and results are based on associations. The importance of tracking changes over time comparing the original and the modified versions, so that responsiveness to clinically relevant changes can be

compared with the 2 versions, will be forthcoming. Results on the basis of cross-sectional data do suggest that increases in precision and the distributional properties of the modified version will result in greater responsiveness to change (Guyatt et al., 1986; Ware, 1987). Second, this is a veteran population of ambulatory users with substantial disease (Au et al., 2001; Fan et al., 2002). Results, while generalizable to other populations of this type, may not be generalizable to healthier populations. However, the performance of these measures in terms of the variance of the scales is similar to a wide range of populations studied and reported in the literature. Be that as it may, other studies are necessary in more general populations.

The methods for conversion of the RP and RE scales from the modified to the original scales are useful for comparisons with version 1.0 of the SF-36. The conversions shown here are appropriate to work in the VA and, similarly, comorbid populations.

The modified role scales are clearly superior to the original versions. The modified version may become a standard in its own right as we develop more experience with its use. While these modifications to the role scales are reliable and valid, the MOS SF-36 has been extensively used in the civilian healthcare system. The question arises as to whether there is a potential loss in the benchmarking of scores with the modifications presented in this article. The conversion formulas presented in

this article provide a bridge to deriving scores that are comparable to the original version. This provides a means for making comparisons with the original version of the SF-36 for benchmarking purposes. In addition, the VA has surveyed over 2 million veterans nationally, using the Veterans version of the SF-36 (that includes the modifications to the role scales), from 1996 through 2000 for monitoring purposes (Kazis et al., 2004; Perlin et al., 2000). Benchmarking of scores is occurring within the VA system for regional and hospital comparisons.

The results of this study show that the modifications to these concepts add more precision to the assessment of role functioning. Additional work that examines these performance characteristics in other populations and tracks changes over time will provide other important evidence to this work. Based on the evidence reported in this article, the modifications to the role scales strongly argue for adopting these modifications in VA populations and others similar to it.

Separately, based upon 10 years of experience, the SF-36 Version 2.0 has been developed for use in diverse populations, with improvements to the reliability and validity of the SF-36 Version 1.0. Both the Veterans SF-36 health survey and the SF-36 Version 2.0 are important short-form assessments for use in populations as measures of outcomes (Ware et al., 2000).

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Appendix

Modified Role-Physical and Role-Emotional Scales Conversion Formula From 5-point Response Choices to 2-Point Response Choices

Scores are converted on the basis of the following recodes. Items are then summed and brought to a normal standard from 0 to 100.

1. Role-physical scale conversion from 5-point to 2-point response choices

During the past 4 weeks, have you had any of the following problems with your work or other regular daily activities as a result of your physical health?

Recode

- a. Cut down on the amount of time you spent on your work or other activities.

Response choice	Conversion	
	From modified (5 point)	To original (2 point)
No, none of the time	1	110.9
Yes, a little of the time	2	58.5
Yes, some of the time	3	21.4
Yes, most of the time	4	-4.9
Yes, all of the time	5	-6.4

- b. Accomplished less than you would like.

Response choice	Conversion	
	From modified (5 point)	To original (2 point)
No, none of the time	1	112.0
Yes, a little of the time	2	68.8
Yes, some of the time	3	29.6
Yes, most of the time	4	-4.2
Yes, all of the time	5	-6.3

- c. Were limited in the kind of work or other activities.

Response choice	Conversion	
	From modified (5 point)	To original (2 point)
No, none of the time	1	108.1
Yes, a little of the time	2	51.6
Yes, some of the time	3	16.9
Yes, most of the time	4	-5.6
Yes, all of the time	5	-6.9

- d. Had difficulty performing the work or other activities.

Response choice	Conversion	
	From modified (5 point)	To original (2 point)
No, none of the time	1	110.7
Yes, a little of the time	2	58.1
Yes, some of the time	3	21.1
Yes, most of the time	4	-5.1
Yes, all of the time	5	-6.6

2. Role-emotional scale conversion from 5-point to 2-point response choices

During the past 4 weeks, have you had any of the following problems with your work or other regular daily activities as a result of any emotional problems (such as feeling depressed or anxious)?

Recode

- a. Cut down on the amount of time you spent on your work or other activities.

Response choice	Conversion	
	From modified (5 point)	To original (2 point)
No, none of the time	1	111.4
Yes, a little of the time	2	58.3
Yes, some of the time	3	12.6
Yes, most of the time	4	-17.6
Yes, all of the time	5	-3.0

- b. Accomplished less than you would like.

Response choice	Conversion	
	From modified (5 point)	To original (2 point)
No, none of the time	1	118.2
Yes, a little of the time	2	62.6
Yes, some of the time	3	17.2
Yes, most of the time	4	-11.7
Yes, all of the time	5	2.1

- c. Didn't do work or other activities as carefully as usual.

Response choice	Conversion	
	From modified (5 point)	To original (2 point)
No, none of the time	1	112.4
Yes, a little of the time	2	60.0
Yes, some of the time	3	14.3
Yes, most of the time	4	-16.3
Yes, all of the time	5	-1.5

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